

The NoSQL world is one without relations, with pure scalability and no joins. NoSQL databases manage data that is not rigorously relational and tabular, so do not use SQL for data manipulation. NoSQL databases are typically non-relational, horizontally scalable, open source, and distributed. A key advantage of NoSQL over SQL databases is its ability to scale an application to new levels. NoSQL databases characteristically highlight horizontal scalability by partitioning and leveraging the elastic provisioning capabilities of the cloud. The NoSQL data services are based on scalable architectures and built for the cloud environment. They provide freedom to select a data model as per needs and use familiar tools.

With NoSQL databases, data replication can be done more easily than with SQL databases. As NoSQL databases are built without relations, data need not be on the same server, and can be processed independently, which allows better scaling than SQL databases. Don't forget, scaling is one of the primary characteristics in Cloud Computing environments. The traditional RDBMS may not be a good fit for cloud-scale applications, due to the strict and upfront schema requirements.

Open source NoSQL databases

The NoSQL movement can be defined by a simple principle: To use the solution that best fits the problem and suits the objectives. Use a key-value-pair database if the data structure is more appropriately accessed through this. And if you have connected data such as social networking or financial transaction graphs, then graph databases are appropriate.

Many cloud applications demand availability, speed, and fault tolerance over consistency, and hence have expanded beyond RDBMS technologies, resulting in the growth of NoSQL databases.

Tabular/ columnar data stores

Tabular/ columnar data stores look similar to tabular databases. Their primary data retrieval model uses column filters by leveraging hand-coded map-reduce algorithms.

HBase

HBase is based on Google BigTable and is column-oriented, open source, and distributed. It uses the Hadoop infrastructure (Zookeeper as a lock service and NameNode, the HDFS file system) and hence supports fault tolerance and scalability inherently, and adds random read-write capability.

HBase tables are distributed as regions, and regions are automatically split and redistributed as data grows. It supports linear and modular scaling, adding RegionServers that can be hosted on the public cloud. Regions are vertically divided by column families into stores, which are stored as files on HDFS. Potential use cases and features include:

- Reads, supported by a single-write master
- Ordered partitions that support efficient row scans

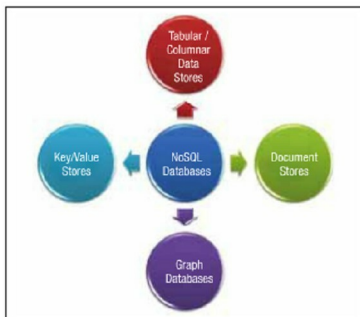


Figure 1: NoSQL databases

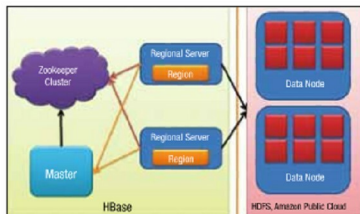


Figure 2: HBase cluster architecture

- Range-based scans
- Batch analysis
- Large caches

HBase does not have many features such as triggers, secondary indexes, etc.

Hypertable

Hypertable is an open source database inspired by publications on the design of BigTable. It runs on top of a distributed file system such as the Apache Hadoop DFS, GlusterFS, or the Kosmos File System (KFS). It is written almost entirely in C++, for performance.

Its features are:

- Scalability: It is based on a design developed by Google to meet scalability requirements.
- Performance: It offers a responsive user experience with low request latency.
- Supports a wide range of applications: Data is sorted by a primary key.
- Cost saving: It has a high capacity on a tiny proportion of the hardware.